LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Description purpose typical pneumatic system components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Description purpose typical pneumatic system components
5. Specific Objectives	23.1.4.1 Trainee will be able to describe the construction and functioning of a typical pneumatic system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system schematic diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of power for pneumatic systems?
(b) Name one common application of pneumatic power in an aircraft.
(c) Why is compressed air preferred over hydraulic fluid for certain systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic principle of pneumatic systems • Construction of air compressor and air turbine • Function of air reservoirs and pressure regulators • Role of air starters and bleed air valves • Safety considerations in high-pressure air systems	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of air compressor operation • Functioning of air turbine and bleed valves • Integration of reservoirs and pressure control units • Operation of air starters and ground power units • Maintenance requirements for pneumatic components	L & Ds	PPT, PDF	20 min
(c)	• Typical layout of pneumatic system in aircraft • Interconnection of components and flow paths • Troubleshooting common pneumatic system faults • Impact of system failures on aircraft performance • Standard operating procedures for pneumatic systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of an air reservoir in a pneumatic system.
(b) Describe how an air starter utilizes compressed air to start an engine.
(c) Identify three safety precautions when working with high-pressure pneumatic systems.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between mechanical, power-assisted, and fly-by-wire control systems to be discussed in LP 20.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min